

Cite as:

Pietrak K, The foundations of socionics – a review, Cognitive Systems Research 47 (2018) 1-11.

DOI/10.1016/j.cogsys.2017.07.001

The foundations of socionics – a review

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Abstract

This paper discusses the foundations of the theory of socionics, including the rationale behind Aušra Augustinavičiūtė's model of information processing by the human psyche, and the philosophical thought behind the concept of information elements. Socionics was developed in the former Soviet Union and may be regarded as analogous to the Myers-Briggs Type Indicator typology of personality. The main goal of this paper is to present socionics to the English-speaking community, in order to better take advantage of its explanatory potential in the fields of interpersonal communication and psychological disorders. Reviewed topics include aspects of Jungian analytical psychology and Kępiński's information metabolism theory, upon which Augustinavičiūtė based her model. After the model is presented, its theoretical implications are briefly discussed.

Keywords: cognitive psychology; biological cybernetics; socionics; information metabolism

1. Introduction

In the 1980s, Lithuanian sociologist Aušra Augustinavičiūtė wrote several papers collected and published in 1998 in a book entitled "Socionics. Introduction" (Augustinavičiūtė, 1998). In these writings, she laid down the foundations of a new theory regarding the processing and exchange of information by human beings. The name given to this new model was **socionics**. As explained by Augustinavičiūtė, it is a combination of two words – "society" and "bionics". The science of bionics is understood to mean examining biological phenomena, in the hope of gaining insight and inspiration for developing physical or composite bio-physical systems in the image of living organisms (Vincent, Bogatyreva, Bogatyrev, Bowyer, & Pahl, 2006). Indeed, the model proposed by Augustinavičiūtė has the characteristics of a cybernetic system although it is built upon the analysis of interactions between living organisms (especially humans) with their environment.

The work of Augustinavičiūtė is based on the findings of Freudian psychoanalysis (Freud, 1915, 1920, 1923), Jungian analytical psychology (Jung, 1946) and the concept of information metabolism devised by Polish psychiatrist Antoni Kępiński (Kępiński, 1972a, 1972b, 1977a, 1977b, 1979). Using unique insights from Kępiński and her own studies of

family relationships, Augustinavičiūtė was able to add details to the functional model of the psyche proposed by Carl Jung in his famous work on psychological types. In this way, a highly systematic description of information processing by the psyche came into being, together with a comprehensive typology of personality types and theory of social interactions.

In the West, the work of Jung evolved into the Myers-Briggs Type Indicator (Myers, 1962, 1980), which became a very popular tool for psychometrics in business and personal development. Since the 1980s, the theory of socionics has been developed in the former Soviet countries, especially Russia and Ukraine. The interest in Augustinavičiūtė's works resulted in the establishment of two socionic institutes (*International Institute of Socionics* located in Kiev and *Scientific Research Socionics Institute* in Moscow). Many socionic experiments were conducted even before the formation of larger research centers, by the creator of the theory and her colleagues from medical and scientific circles. These experiments usually involved the determination of psychological types (using questionnaires and interviews) for a sample group, followed by the observation of their interactions. The results of some of these experiments were published in Russian (Reinin, 2005). The socionic model offers much more than MBTI typology, as it allows us not only to specify the psychological type of the individual, but also to predict problems in their interactions with others and society as a whole (Augustinavičiūtė, 1998). Horwood and Maw demonstrated that socionics may be used to increase team cohesiveness in operating theaters and other high-stress workplace environments (Horwood & Maw, 2012). In their study, the performance of theater teams working in Glan Clwyd Hospital (Wales) was optimized by relocating certain employees to achieve greater psychological compatibility between the members of each team. Reinin investigated psychological and somatic effects of short and prolonged interactions in groups made up of different combinations of socionic types (Reinin, 2005). He reported that certain arrangements resulted in positive health effects (normalization of blood pressure and pulse for all participants) while others brought about negative consequences, such as increased mental strain and nervous breakdowns. Eglit applied the socionic model in her clinical studies of personality disorders. She found that these types of disorders can be explained within the framework of the theory of information metabolism distortions (Eglit, 2013).

Despite its immense potential, socionics has failed to evoke greater interest among the Western psychological community. The comparison of Google trends for searches related to two topics – socionics and MBTI – is shown in Figure 1. Interest in MBTI is on average 7.6 times greater than in socionics, yet the latter dominates in the former Soviet countries. The popularity of MBTI typology can be related to its simplicity. Models of types and diagnostic tests are much less complex within the MBTI system. Another probable cause for the prominence of MBTI in the West is the language (and, in the past, political) barrier. The original writings of Aušra Augustinavičiūtė and her colleagues are available only in Russian, and some of them have not yet been digitalized. The osmosis of these concepts through the iron curtain was highly suppressed. Another distinct aspect of socionics is its philosophical and scientific foundation, which often seems unclear to Westerners.

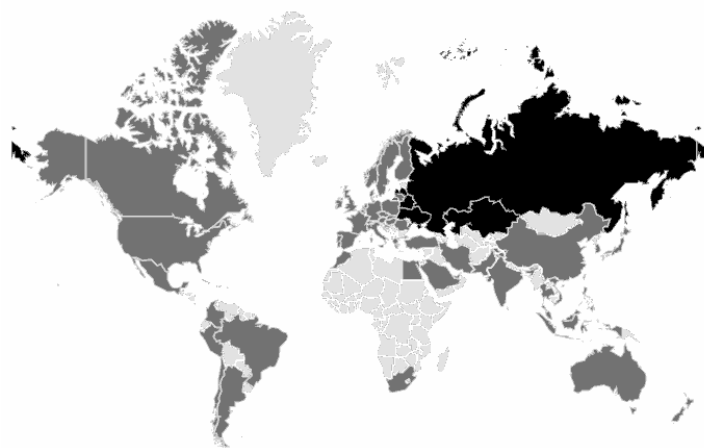
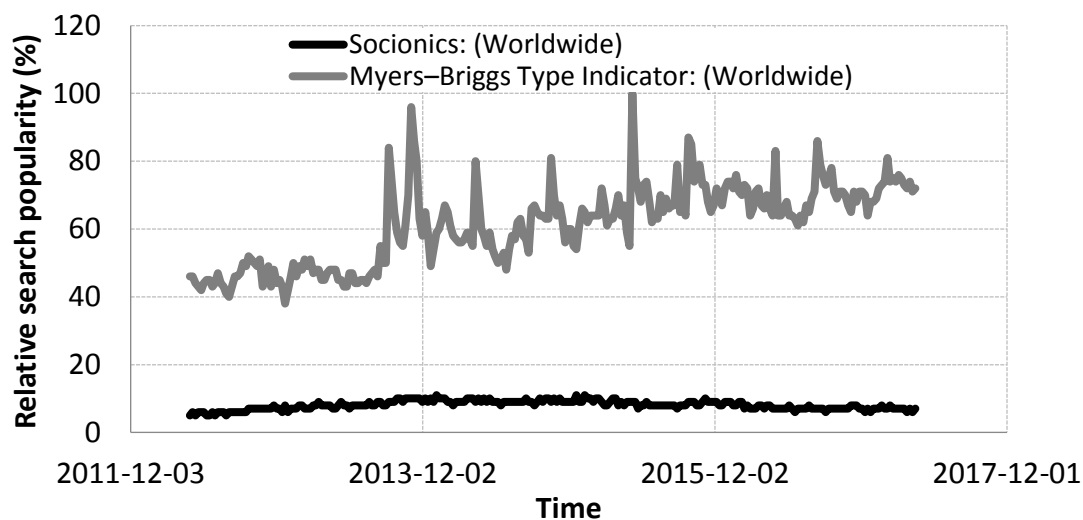


Figure 1. The comparison of popularity of Google searches for two topics, Socionics and MBTI, in the last 5 years, generated via the Google Trends website (top). Interest per world region (bottom; socionics – black, MBTI – dark gray). Although the worldwide interest in MBTI typology is on average 7.6 times greater than the interest in Socionics, the latter system dominates in Google searches in the post-Soviet countries.

This paper addresses two of the aforementioned problems. Its first aim is to introduce Augustinavičiūtė's philosophy, which forms the foundations of her model, to the English-speaking community. The second aim is to provide clarification of some intricate and commonly misunderstood aspects of her system. The paper starts with a compact formulation of Jung's functional description of the psyche, as the terms devised by him are crucial for the understanding of the socionic model. These terms are discussed together with the basics of Kępiński's information metabolism in section 2. In section 3, the concept of information elements, which is a direct result of the combination of two theories introduced in section 2, is discussed. Chapter 4 deals with the derivation of model A – the model of the human psyche first put forward by Augustinavičiūtė. The final sections contain a discussion and summary.

2. Prerequisites

2.1 Jungian functional model

The starting point for socionic considerations is Carl Jung's concept of the 4 functions of the psyche (Jung, 1946). Jung built his understanding on Sigmund Freud's theories of the conscious and unconscious mind. In Jung's view, the activities of the mind could be described by the functions **sensing, intuition, thinking** and **feeling** (brief descriptions of these functions are given in section 2.2). From the analysis of his patients and various historical figures, Jung concluded that usually one of these functions is much more prominent in an individual (by unconscious, not purposeful, selection) and better developed than the others. He regarded this one-sided development as an effect of the individuals's adaptation to their environment. To refer to the most prominent function, he used the term "**dominant function**". The dominant function is highly conscious, unlike the three non-dominant functions. Additionally, he identified one **auxiliary** function – less conscious than the dominant one, but acting as its main support. He also inferred that because of one-sided development (with the focus on the dominant function), one or two functions are necessarily impeded in their development, and can be called **inferior**. In his view, an inferior function (or functions) is placed very deep in the unconscious – the person can be unaware that these kinds of perception or decision-making are possible, which does not mean that these functions are not affecting their overall behavior. These functions continue to act but remain hidden (Blutner & Hochnadel, 2010).

Based on Jung's observations, Isabel Myers suggested a more systematic functional model (Myers, 1962, 1980). She proposed that there is only one highly unconscious **inferior function** and a **tertiary function** is moderately unconscious and developed somewhat better than the inferior. Jung's original reasoning was more flexible. He emphasized that some universal principles and tendencies are visible in the structures of different psyches, but there was room for exceptions and deviations from the general scheme. For example, in some individuals, none of the functions is particularly prominent (they lack a dominant function). Disorders in the functional structure of the psyche were seen by Jung as the source of psychological problems. In general, he stated, a person is naturally set up to possess one specific functional type, either sensory, intuitive, thinking or feeling. If another type is forced upon a person due to external influence, he or she is likely to become neurotic in the future, and effective treatment is possible only through identification of the proper dominant function.

Carl Jung also identified two preferences concerning the source of psychic energy (the psychological force which gives us the drive to do something; Freud called it the *libido* and associated it with sexual motivation, but Jung extended Freud's understanding of it and added other sources, like the need for sacrum and transcendence). Psychic energy can be acquired from the external world or from the inner world of an individual. If the energy level of the psyche increases due to external motivation, e.g. dealing with other people and objects, this direction of energy flow is called **extraverted** (from Latin: *extra* – outside, *vertere* – turning; "turning outside"). Conversely, if the level of energy increases

as a result of dealing with one's own sensations, intuition, thoughts and feelings, this kind of energy flow is called **introverted** (from Latin: *intro* – inside, *vertere* – turning). It should be noted that the psychoanalytical meaning of energy used by Freud and Jung should be considered metaphorical, not physical. As Jung suggested, the individual usually prefers one source of energy, inner or outer, while the other is suppressed. These two orientations can be paired with the 4 mental functions, to give two versions of each function – extraverted and introverted, respectively. This reasoning yields 8 functions. Bearing in mind that in people, one function is typically more developed than others, 8 personality types can be distinguished (Jung, 1946).

The types may be broken down further into 16 types if the auxiliary function is considered. In Jungian personality theory, this function is usually considered conscious, while tertiary and inferior functions are seen as unconscious (Pascal, 2009). Isabel Myers interpreted Jung's writings and came to the conclusion that the auxiliary function must be opposite in direction to the dominant function (Myers, 1962, 1980). One can infer intuitively that it is necessary for a person to have both introverted and extroverted functions in the conscious part of the psyche, as without this the person would not be able to consciously gain energy from both sources – internal and external. When the dominant and auxiliary functions of the person are determined, the position of the two other functions is also determined. The inferior function must be opposite to the leading function. If one direction of energy flow is developed (with the development of the dominant function), then the other one is neglected. Therefore, an extroverted type has an introverted inferior function, and vice versa.

It should be noted that cognitive functions are paired according to their task. Sensing and intuition are two functions whose task is to make observations about the world. Importantly, as emphasized by Jung, they do not overlap. This means that observations made by intuition are of a completely different nature than those made by sensing. The same can be said of evaluations made by thinking and feeling. The focus on the leading function and its development results in the suppression of the other function of the pair. It means that a type with a leading sensing function has inferior intuition, and vice versa. By the same principle, developed thinking results in inferior feeling. Additionally, it is also logical to assume that if one of the conscious functions makes observations (sensing or intuition), the other one should make evaluations (thinking or feeling). Only in this kind of configuration will there be potential for mutual support.

2.2 Jung's understanding of the 4 functions

In Jung's view, the psyche makes observations of reality only through sensing and intuition. Sensing relies on the senses. It only shows that something is present – an image, a sound, a smell, a taste, a bodily sensation – and makes no further mental operations upon these qualia. Intuition also allows us to know that something is, without raising conclusions or reflecting (as these are the tasks of thinking and feeling). If one knows something which is not confirmed by a sensory experience, this is intuition. The source of such a belief is not explicit. Thinking and feeling are used to operate on observations made

by intuition and sensing. Thinking is the sharp blade of logic. It allows us to differentiate between elements, compare them, identify patterns and repetitions, causes and effects, and formulate relevant judgments. It uses cold analysis, without the consideration of feelings. The principles upon which the feeling function makes its judgments are of a fuzzy nature. The premises are not clear. Frequently, the verdict of feelings is illogical, yet a feeling type relies upon it.

Let us emphasize an important quality of Jung's model. The functional division of the psyche is complete. All activities of the psyche are manifestations of one of the 4 functions – there cannot be an activity which does not belong to one of them. Additionally, any activity of the psyche is characterized by an extraverted or introverted attitude.

2.3 Kępiński's information metabolism

Carl Jung's functional description of the psyche was the basis upon which Aušra Augustinavičiūtė constructed her system, describing the processing of information metabolism by an individual psyche as well as the complexity of interactions between multiple individuals. The idea of considering the psyche as an information processor of stable structure was inspired by works of Polish psychiatrist and philosopher Antoni Kępiński. In his books, Kępiński introduced the notion of **information metabolism (IM)** (Kępiński, 1972a, 1972b, 1977a, 1977b, 1979). In brief, he reasoned that, as a living organism metabolizes matter (which he referred to as energy elements), while retaining a macroscopically unchanged structure, so the psyche metabolizes information, while the overall structure of the psyche remains unchanged. By **energy metabolism (EM)**, he was referring to accepting food and oxygen from the outside world and expelling metabolic waste products. In case of information metabolism, the individual reacts to perceived information, and this reaction acts as feedback to the outside world. The life of an organism is possible if both EM and IM are occurring, and it stops if one or both are ceased.

In his reasoning, Kępiński utilized the concept of negentropy (the measure of order) in the same way as Brillouin, who proposed a link between the thermodynamic and informational interpretations of that term (Brillouin, 1950, 1953). It was Erwin Schrödinger who devised the idea that biological entities are different from inanimate objects in the sense that they do not obey the principle of increasing entropy, but rather increase their own negentropy at the expense of the negentropy of environment (Riedl, 1992). Non-living objects merge with their surroundings while living organisms form separate, ordered entities. Bertalanffy considered these entities open systems, and proposed their formal description (von Bertalanffy, 1950).

The tendency to increase negentropy is brought about by biological organisms in multiple ways. It manifests itself in the protection of their physical (body) and mental (psyche) structures from dissolution, the organization of their immediate surroundings according to their needs and the passing on of their genetic information to future generations. In other words, an organism wants to preserve its IM, and moreover, to imprint it onto its environment (Kępiński, 1977b, 1979). Kępiński viewed these two objectives as the

superior governing principles for biological entities, which also apply to humans. In favorable material and psychological conditions, these imperatives may be effectively followed. In the presence of serious obstacles, the metabolism of information is inhibited or interrupted, which leads to a variety of mental conditions. As noticed by Kępiński, the general dynamics of vital processes is strongly correlated with mental health (Kępiński, 1979). Healthy individuals readily engage in increased exchange of information with the environment, e.g. meeting new people, exploring new places and trying new activities. By contrast, depressed patients tend to hide from the world and minimize their chances of new experiences and their overall activity.

If an organism has limited access to energy resources (e.g. food and life space), its IM drops (there is less energy for information processing and work). On the other hand, there are psychological factors which also lead to the disruption of IM. These factors were investigated analytically by Kępiński, on the basis of his two-phase model of information metabolism, described most elaborately in “Melancholy” (Kępiński, 1979). A short description of the two phases of IM is given below.

The transfer of information is initiated by the perception of change in the internal or external environment of the biological entity. This change is registered and transmitted through the system in a purely cybernetic manner, which agrees with the intuitive definition of information given by Bateson (Bateson, 1972):

What we mean by information — the elementary unit of information — is a difference which makes a difference, and it is able to make a difference because the neural pathways along which it travels and is continually transformed are themselves provided with energy. The pathways are ready to be triggered. We may even say that the question is already implicit in them.

The first phase of IM begins after the aforementioned initialization. It has the characteristics of instinctual and emotional evaluation of the perceived change. The individual formulates an initial attitude towards the phenomenon, based on the subjective feeling created by the event. The person usually has little control over that feeling, therefore it can be categorized as subconscious. There are two basic attitudes the psyche may adopt – “for” or “against”. In healthy individuals, dangerous events induce the attitude 'against', and positive ones – 'for'. In unhealthy individuals, this classification may be distorted. After the formulation of the initial attitude, the second phase of IM may begin. In this phase, the reaction is carried out. The psyche chooses the most appropriate action from many known procedures and reality maps (referred to as functional structures). At this point, there is also the possibility of distortions and blockades contributing to mental disorders. The description of mental dynamics as a conceptual grid, proposed by Kępiński, was developed further by various authors. Struzik discussed the concept of Kępiński's functional structures and their connection with the topological networks of Kohonen (Struzik, 1986a). Within this paradigm, he proposed explanations of aphasia, apraxia and catatonic schizophrenia. In another paper, Struzik proposed a

further generalization of Carnot's principle given by Brillouin (Brillouin, 1950, 1953) using the concepts of energy and information metabolism. Brillouin's model was extended to include information stored in the brain, which was omitted from the original model. This conceptual framework was used to explain the differences between decreases in IM rates observed in depressive and early-schizophrenic patients (Struzik, 1986b). Kokoszka continued the development of the IM model by drawing on it to develop the theory of states of consciousness and protective mechanisms of information (Kokoszka, 1993, 1999). A description of the IM model in mathematical terms was attempted by Kokoszka, Bielecki and Holas (Kokoszka, Bielecki, & Holas, 2001). A comprehensive summary of all works related to the IM model mentioned so far is given in a book by Kokoszka (Kokoszka, 2007). More recently, Bielecki reviewed cybernetic descriptions of the phenomenon of life, including many ideas from Kępiński (Bielecki, 2015).

3. Information elements - the fusion of Kępiński and Jung

Everything that happens in observable reality, including our own bodies, may be reduced to energy changing its forms. Here, energy is understood in the strictest physical sense. The equivalence of matter and energy was proved by Einstein and therefore matter is not considered a separate category. Although Kępiński used the notion of energy metabolism (EM) only in association with the dynamics of the life processes of a biological organism, Augustinavičiūtė used it to describe energy conversions in general. The latter convention will be used in this section, where the derivation of **information elements** (a novel concept developed by Augustinavičiūtė) will be presented.

Let us assume that the psyche is observing an event. Any transformation of energy – a unit act of EM – is accompanied by the generation of some information (e.g. if an object moves, explodes, collides with another object – we notice the change). Augustinavičiūtė chose to define the units of information metabolism as corresponding to specific observable transformations of energy. To find out what these energy transformations might be, an object that is representative of all existing objects should be considered. The consideration of inanimate objects only is too narrow, as the psyche deals with much more complex objects, such as people and their emotions and needs (as well as its own). Therefore, an analysis of energy transformations during the activity of the human body would be appropriate. The founder of socionics compared living organisms to an internal combustion engine and to a bouncing elastic ball (Augustinavičiūtė, 1998). Before the presentation of these analogies, a short, clarifying introduction will be given.

One should notice that organisms are frequently treated as open thermodynamic systems and such a description is quite common in engineering (Cengel & Boles, 2006). The internal energy of an open thermodynamic system may be altered by transfers of enthalpy, heat and work. For biological organisms, the contribution of enthalpy transfers to the overall energy balance is very small and can be neglected. Heat is released during the numerous chemical reactions taking place in the cells. In these reactions, nutritional components are metabolized, therefore we can think of food as fuel. Let us notice that heat

is a disorganized form of energy, associated with chaotic motions of particles. Uncontrolled heat transfer leads to the thermal death of the system – the state at which it can no longer perform any work. By contrast, work is an organized form of energy in which particles are moved in an orderly way, in a specified direction (consider a body pushed by a piston). This motion can be directed to desired locations through mechanical links, to perform various tasks. In the case of a living body, we can think of blood circulation, breathing, peristalsis and motor functions in general.

A device which uses heat to generate work is basically a heat engine. This type of modeling was used by Augustinavičiūtė regarding organisms. She divided the general activity pattern of an organism into 4 stages. Firstly, let us observe that any activity requires some form of potential energy. This energy is contained in the structure, like the chemical energy stored in fuel, or the potential energy of elasticity contained in a squeezed rubber ball. Augustinavičiūtė assigned the symbol (▲) to this energy state. The next stage is the release of potential energy from the object and its conversion into kinetic energy – the process of mobilization. This stage is dynamic and includes a change in the internal structure of the object. It corresponds to the compression of the fuel-air mixture, or the return of the elastic ball to its unsqueezed shape. This dynamic act of EM was assigned the symbol (▣). At some point, the object reaches its moment of maximal mobilization, where it has the greatest kinetic energy. This corresponds to the moment of ignition in an internal combustion engine, or the ball state just after decompression, and at the start of its upward motion. This phase of EM is assigned the symbol (●). The last stage of this cycle is work. The combusted mixture expands and is able to move the piston. The ball can overcome the Earth's gravitational pull and jump off the floor. As work is done, the kinetic energy of the body decreases. This is the dynamic stage and is assigned the symbol (■).

Humans and animals also act according to the aforementioned stages. If the person is at rest, there is the potential of action (▲) which can be realized if they get excited by a stimulus. The process of mobilization (▣) may be linked to the first phase of IM mentioned by Kępiński, in which emotional evaluation of the stimulus takes place. It invokes an attitude 'for' or 'against' the perceived change. The body is rapidly prepared to execute an adequate physical reaction, for example to escape or to draw nearer to the source of the stimulus. At the beginning of the execution of the physical reaction, it possesses maximal kinetic energy. This state is referred to as (●). To accomplish the selected goal, the body must expend energy. In physics, this is called work (■). This process also requires thinking – for example planning an appropriate sequence of actions or applying tools. In the process, the person tires. Please note that these are phases of EM, but by observing them, the psyche learns and obtains information. It means that each stage of EM mentioned is accompanied by information content characteristic for that stage. Each change in the body itself or in its position is not only an act of energy metabolism, but also an informational signal to surrounding living beings. This understanding of the link between EM and IM was accentuated by Augustinavičiūtė.

Let us notice that Augustinavičiūtė divided the EM process into phases which can be easily linked to the 4 Jungian functions. As a matter of fact, if the Jungian paradigm is adopted, the information about potentials of objects is related to imagination and acquired by intuition, specifically by its extraverted version (Ne). The information about releasing this potential - mobilizing the body into action, is related to extroverted feeling (Fe). Consequently, the information about the object in its full explicit power is obtained by extroverted sensing (Se), while information about work done by objects in the physical world is obtained through extraverted thinking (Te). There is one apparent inconsistency, as Jung pointed out, that the input into the psyche is only via sensing and intuition, and not by thinking or feeling. Indeed, thinking and feeling work on the stimuli received through senses and intuition, but, here, the information obtained through Te or Fe is considered something which, of course, has its basis in sensory inputs and intuitions, but is focused more on further reflections than raw input data.

For completeness, it is necessary to discuss the introverted side of the 4 functions, as only their extraverted versions have been mentioned thus far. As noticed by the founder of socionics, introverts think not in terms of objects and what happens to them, but in terms of relationships between objects, especially the relationship between the subject and the external world. To refer to relationships, Augustinavičiūtė utilized the concept of a field, taken from theoretical physics. This association should be considered metaphorical and can be easily explained. Celestial bodies interact mostly from afar due to fields. The field is a sum of relationships of one body with other bodies. It becomes clear if we imagine a particle. We cannot say if it has an electric charge (and therefore an electric field around it) unless we put another charged particle nearby and see how it reacts. It may move closer, drift away or stay in place. Only then can we say something about the relationship between the charges of these particles. Other physical fields work in the same way. Therefore, the concept of a field is somehow akin to the concept of a relationship. Now, let us consider that any transformation of the energy of an object – e.g. motion, excitation, deceleration, work – is accompanied by change in the surrounding fields. An object might have changed its position in space, so its spatial location with respect to other objects may have changed. The person might get excited or depressed, and thus their energy level will change relative to another person or people. In the last example, the term “energy” was used in the psychological sense, but it should be kept in mind that behind all activities there are physical processes of energy conversion.

In the previous paragraph, it was demonstrated that every element of EM (and IM) referring to objects can be assigned a corresponding element related to fields. Among object-elements, there are 2 static and 2 dynamic elements. In the Jungian model, the functions are characterized by symmetries, so symmetry is also expected between object- and field-elements. The first object-element is potential energy (▲) and it is static. The field version of this element is denoted by a white triangle (△), and it refers to the dynamics of the field of potential energies of objects. Potentials possessed by various objects may unfold in the future. This element is associated with time, as time is the

manifestation of events which had previously been mere potentials. This element is dynamic as it describes the process of unfolding, and relational as it describes the field. It corresponds to Jungian introverted intuition (Ni).

Now let us think of the field version of element (■). It will be denoted by white symbol of the same shape (□). The process of internal mobilization of the object is dynamic, therefore the field-element associated with it should be static, by the law of symmetry. Living organisms become motivated to action due to the feeling of attraction (or repulsion) to something. Such a feeling of attraction or repulsion can be seen as a static moment (still snapshot) of (■). Therefore, (□) is the static field of attraction/repulsion (between all objects, with the emphasis on the place of oneself in it). The explicit state of energy is (●). Its field-element should be dynamic and is denoted by a white circle (○). It can be described as the dynamics of the relationship between explicit energy states of objects. If Jungian functions are considered, it is related to introverted sensing (Si).

The field-element associated with work is related to Jungian introverted thinking (Ti). As work (■) is dynamic and requires purposeful procedures, the information about explicit statics of relationships between objects (□) is needed to avoid chaotic (and therefore useless) actions. Treating objects as aggregates of static (frozen) properties allows for their comparison and classification, which helps in making decisions and building mental models of reality (Kepiński, 1977b).

As before, any field-element of EM implies the existence of an associated IM element. In summary, Augustinavičiūtė classified the information received by the psyche into two categories:

- (1) sensations that are formed into perceptions and bring information about the qualities and condition of objects (**black elements of IM**);
- (2) sensations that are formed into impressions and bring information about fields (**white elements of IM**).

The quality of received information depends on the characteristics of external stimuli and the quality, strength, development, and training of the personal information-receiving system. At this stage it is worth mentioning that various authors have given slightly different definitions of IM elements. Even Augustinavičiūtė's definitions changed from paper to paper, yet still the qualities which these descriptions tried to convey were always the same. In some socionic papers, descriptions of IM elements involve many examples, which is good for beginners and people with weaker intuition (Eglit, 2010). At the same time, these kinds of descriptions can be too specific, and the necessary level of generality may be lost.

4. Model A

4.1 Introductory considerations

When IM elements are defined, one can think of the way in which the psyche processes information. Augustinavičiūtė used Jung's observations again and reasoned that one of the elements must be processed by some sort of dominant function. There must also be a second, auxiliary element, of a different type. If the dominant function works on a black element (objects), the auxiliary function must work on white (fields), and vice versa. This is in accordance with the observation that a change in the object causes an immediate change in field. The psyche is likely to function in this way – first observe, then adjust something, or first adjust, then observe reality's reaction. It is also understandable that if that one of two major functions is sensing or intuition, the other should be feeling or thinking (and vice versa, just like in the MBTI system). In conclusion, Augustinavičiūtė proposed that there are blocks in the psyche, composed of one white and one black element. Freud divided the psyche into ego, superego, and id (Freud, 1915, 1920, 1923). These names were her initial choice for the names of functional blocks. Augustinavičiūtė knew the writings of Jung, who argued that the psyche acts as two separate systems or relatively independent personalities – the conscious and the unconscious (Pascal, 2009). She also kept in mind that conversions of energy in the material world form cycles. Her conclusion was that the psyche is formed of two separate cycles of information metabolism – conscious and unconscious. As all activities of the psyche are described by Jungian functions, therefore these cycles should contain all 8 IM elements. There must also be space for the entities identified by Freud as the ego, superego and id.

4.2 Mental ring

The conscious cycle may be built starting from elements placed in the **dominant** and **auxiliary functions**. Assume we have an individual with dominant (●) and auxiliary (□). These form one block of IM, which corresponds to the Freudian **ego** – the conscious and strong foundation of a personality. Another thing which can be relatively easily brought into the conscious field is the super-ego. **Super-ego** (meaning *the one that supervises ego*) reflects the internalization of information forced upon an individual by society, the cultural rules. It is often taught by parents and educators. People typically use their learned norms when a problem involves one of their weaker psychological functions. In such a situation, it is easier to refer to social norms on a given subject than to form a solution of our own. Augustinavičiūtė concluded that this block is weak and formed of the remaining two IM elements, that have the same dynamics as the elements belonging to the ego block. In our example, the ego block contains two static elements (●□), in which case the super-ego must also be static (□▲). In this way, the conscious cycle of IM metabolism was specified. It is referred to as the **mental ring** and its graphical representation for one of the 16 possible arrangements is given in Figure 2.

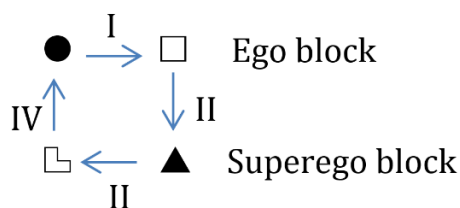


Figure 2. An example of conscious mental ring information metabolism in the psyche for type ●□.

The full cycle of IM consists of the two blocks – ego and superego, and the four phases I-IV. In each phase, we must consider how the first element influences the next one (following the direction of the arrow). Augustinavičiūtė named the first element in each block **accepting**, as this is the function which primarily accepts information from the outside world. With this function, one perceives the state of reality. This perception is so strong that it is sometimes experienced as fusion or union with the reality being reflected. In the ego block, the dominant function is accepting. The other function in the block is referred to as **producing**. Producing functions are also able to accept information about reality, yet their primary focus is creative – they formulate the response to the image of reality perceived by the accepting function. For example, in block ●→□ the effects of the properties of the object (●) on the relationships of other objects to it are processed. To change its relationship with other objects, the properties of the object must be changed. This kind of block is called **extraverted** (due to the type of the element in its first function). In an **introverted** block, e.g. □→●, how the relationship of an object to other objects affects the object itself is processed. The first element is taken as given, and the second one as resulting from the first.

These principles manifest in real people. Let us consider individuals of type ●□ (types in socionics are denoted by elements processed by their dominant and auxiliary functions). These individuals hold the view that the strongest person sets the rules (the qualities of the object are taken as given, and the relationships as resulting from these qualities). The introverted type □● is different. Her/his ego block processes how relationships affect the object. For example, if such a person holds the position of an ordinary teacher, she/he is quiet, modest and scrupulous. Chosen to be the principal, she/he is likely to become dictatorial (Augustinavičiūtė, 1998).

The concept of accepting and producing functions is clear if we consider the cybernetic view. The simplest cybernetic system should include a detector, a signal processing unit and an effector or action-taking subsystem. In case of the functional block, the accepting function acts as detector, the producing function acts as information processor and various parts of the body may be regarded as effectors.

Let us now consider the characteristics of the superego block. The functions of that block are referred to as role function and vulnerable function. **Role function** comes 3rd in the order of processing (in the model shown in Figure 2 it processes the element ▲). Due to the order, it is an accepting function, and due to its placement in the Superego block, it is less prominent than the dominant function. Its name reflects the fact that the individual

sees an element processed by this function as something important from the point of view of others and tries to adjust to these expectations by training this function and demonstrating her or his skills in this area. The need for improvement and a good command of this element can be seen as a role imposed by environment. **Vulnerable function** comes 4th in the order of processing. Its efficiency is even weaker than that of the role function because, as a producing function, it mainly formulates responses using the input of the already weak 3rd function. Unlike accepting functions, which reflect the objective image of reality, producing functions create new qualities, which are not present in the objective world. Conclusions constructed by a weak function from incomplete information delivered by another weak function are likely to be of questionable quality. The correction of mistakes consumes time (and energy). The incompleteness of information input from the external world and frequent mistakes are the reasons why the 4th function is the bottleneck of the mental ring. Information metabolism undergoes a significant deceleration at this point.

4.3 Vital ring

Using the mental ring model, the part of the psyche of which we are consciously aware can be understood. When using these functions, a person can usually control what they say (except for those suffering from mental disorders), and know what they feel. They have the ability to act purposefully. The other cycle of IM in the psyche is unconscious and is known as **vital**, as it works in the background and secures the basic needs of the body (from Latin *vitalis* – life).

Jung described how the unconscious contains instincts, archetypes and complexes (Pascal, 2009). Archetypes act as a sort of software supporting our basic orientation in reality, whereas instincts provide rapid reactions in life-threatening situations to ensure survival. The activity of the vital ring is strongly bonded with the body, while the mental ring is more related to the mind. This unconscious ring contains 4 elements. If all elements of the mental ring are static, then all elements of the vital ring must be dynamic, and vice versa.

The most natural choice to start the unconscious cycle of IM is the inferior function (it will be denoted by number 5). The inferior function should process the opposite version (extraversion/introversion) of the element to the one placed in the dominant function, which is a logical implication of the imbalance caused by the development of the dominant function. Moreover, having sensing as the dominant function implies that intuition is the inferior function (and vice versa). The same is true for thinking and feeling (as explained earlier in section 2.1). For type **●□**, whose mental ring is described in the previous section, the inferior function is (**△**). It should be in one block with a black element (as explained above). The most obvious selection is (**■**), which is the extroverted version of the element located in the vulnerable function. A block built in this way is unconscious and weak. Even if the second function of the block is strong (in this example - the one processing **■**), the overall performance of the block will be degraded by the presence of

the inferior function. In socionics, the inferior function was treated as an accepting function, and given the name **suggestive function**. The other function residing in the same block was called the **mobilizing function** (number 6), and the block itself was named the **superid** (meaning: *the one which supervises id*).

Let us explain these names. As this is the block of unconscious weaknesses, the individual reacts very positively to help in these areas. The level of automation of these functions is high, and capabilities are low, therefore the person can be easily manipulated using information of this type. Comprehensive and accurate judgments relating to information of this type cannot be easily made. Due to these characteristics, this block describes the most childlike sphere of the psyche. The accepting function, which tries to establish union with the objective world, is especially prone to naivety. The term 'mobilizing function' is related to its special placement in the vital ring. It is weak, but when given clear external information, it can induce acceleration in the remaining section of the vital ring (functions 7 and 8, see Figure 3 and Figure 4).

As we have exhausted all other options, functions 7 and 8 must contain the other versions of the elements from the dominant and auxiliary functions. For example, in the case of type ●□, function 7 is occupied by the IM element (○), and function 8 by (■). Information processed by function 7 is usually ignored by the conscious mind, so this function is known as the **ignoring function**. The reason behind this is simply the prominence of the dominant function, which processes very similar information, although it is represented in a different way. Information about objects can be translated to information about relationships between them, and vice versa. Yet the individual avoids such translations if there is no overwhelming need to think in terms of an element processed by function 7. Eglit pointed out that the psyche uses information of this type in situations of self-defense (Eglit, 2010). These uses are highly instinctual and belong to the self-preservation programs of the unconscious. An individual has very little control over these aspects of their psyche, and its responses are very fast. The last function of the vital ring – function 8, or the **demonstrative function** – works with the ignoring function in one block, and therefore it is dependent upon its input. At the same time, as a producing function it is more flexible and less involved in the observation of reality. The element processed by it is qualitatively related with the element occupying the auxiliary function. This means that the information element processed by function 8 is not perceived as important. This type of information is already skillfully handled by the ego block. If other people try to convey this type of information, it is seen by the ego as obvious and even boring. However, on an unconscious level, the mind frequently relies on the output of the demonstrative function and uses it in the personal sphere. This function is often used to neutralize dangers in the environment noticed by the ignoring function.

Let us expand the model of the psyche (referred to as **model A**, after the name Augustinavičiūtė) presented in Figure 2 by adding the elements processed by the functions of the vital ring, following the considerations presented so far. This model for the type ●□ is presented in Figure 3. The general schematic of model A is presented in

Figure 4. The hypothesis of Augustinavičiūtė was that any psyche can be accurately described by this framework. Just like in the MBTI typology, there are 16 possible variants or types resulting from this theory. These variants are constructed based on possible variants of elements arranged into blocks of dominant and auxiliary functions. The dominant function (often referred to as the leading function) may process any of the 8 IM elements, while the elements processed by the auxiliary function are selected by the same rules as in the MBTI system. The location of the 6 remaining elements is then automatically determined, if principles presented in this and the previous sections are followed.

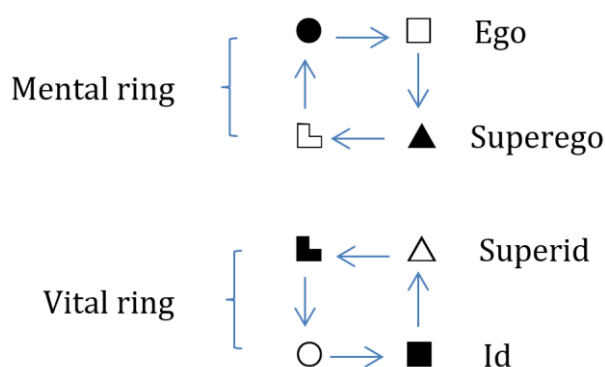


Figure 3. Model A for the psyche of type ●□. Arrows indicate the direction of information metabolism

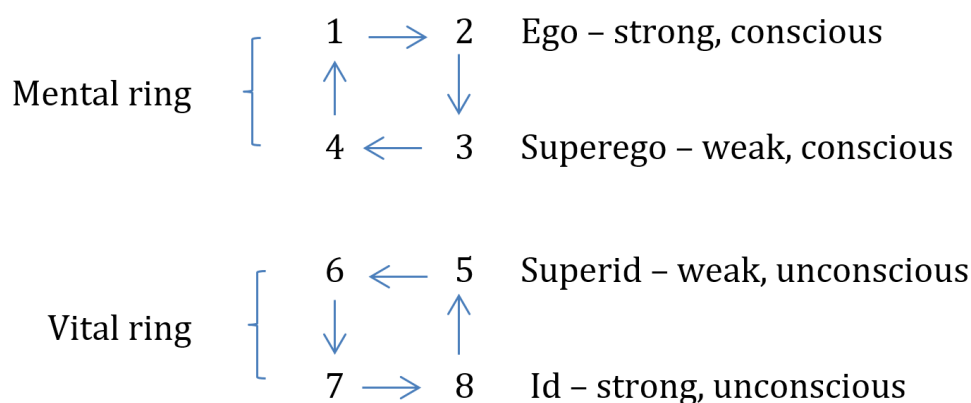


Figure 4. The numbering of functions of Model A. Key to function names: 1-dominant (leading), 2-auxiliary (creative), 3-role, 4- vulnerable, 5-suggestive (inferior), 6- mobilizing, 7-ignoring, 8-demonstrative.

4.4 Basic properties and implications of model A

It is worth mentioning the important role played by weak functional blocks (the superego and superid). As IM is sequential, and the functions are codependent, the level of development of weak functions and the quality of information to which these functions are exposed affect the quality of IM in the whole psyche. Because of that, weak blocks must

be considered control blocks. The bottleneck nature of function 4 in the mental track has already been explained. For the vital track, the real accelerator is function 5 (the suggestive function). Without adequate information input, it can slow down IM in the vital ring. This is also true for function 6. These dependencies can be easily combined with Kępiński's views, in which mental illnesses were seen as disorders of information and energy metabolism in an individual. When his theory is considered, it can be concluded that the socionics model has explanatory potential in the study of mental illness.

Due to the specific roles held by different functions of model A, 16 qualitatively different intertype relations exist. The theory of intertype relationships may be found in the collected works of Augustinavičiūtė (Augustinavičiūtė, 1998). The distinctive quality of that theory is the fact that it can be easily described in terms of mathematics, similarly to other Jungian personality theories, see e.g. (Blutner & Hochnadel, 2010). Socionics predicts that interactions between some agents (types) have a beneficial influence on the metabolism of information in both agents, while there are also relationships which result in the suppression of information metabolism in one or both of them (asymmetrical relationships). Relationships in which IM is suppressed are unfavorable in the light of Kępiński's observations about the connection between information metabolism and mental wellness. Apart from one-to-one relationships, dependencies between agents in multi-agent groups can be predicted.

5. Discussion

Transformations of energy – the changes in the external reality – are acquired by us through all 5 senses. In this respect, material objects manifest not only as visual qualia, but also as sounds, smells, tastes and bodily sensations. Any of the senses can be considered to possess its own space and time dimensions. The socionic model does not describe how these external stimuli are integrated into the information about objects or relationships between them that is later processed in the mind. The focus is clearly on information that has already been pre-processed by lower-level cognitive structures. Due to that, socionics may be considered an extension of analytical psychology. At the same time, socionics offers some hypotheses about the control flow in the psyche. According to them, the efficiency of information processing and even overall mental health can be controlled by exposing weak functions to compatible (or incompatible) forms of information (information elements). This type of consideration is characteristic for the fields of system science and cybernetics. At the most general level, socionics may be classified as part of the cognitive sciences.

The notion of an object used in the descriptions of information elements requires some attention. When we consider what is treated by the psyche as an object and what is not, the hypothesis is that extraverted types (the ones possessing an object-element in the dominant function) tend to pre-process information to give it the form of perceptions, thoughts, feelings and sensations centered on a focal point (which is a broader category than just physical objects with clearly traceable boundaries). On the other hand,

introverts are more likely to initially organize incoming information into information about relationships between these focal points. A concept, a remembered place or an event may be used as a focal point.

The concept of communication or flow of content between conscious and subconscious fields of the psyche is quite important in analytical psychology, therefore it was also considered in socionics. The possible route of information between mental and vital rings was discussed by Gulenko (Gulenko, 1998). This question was also investigated by Yermak, and the results of his analyses were described in his books (Yermak, 2003, 2009). The latter author suggested that the transfer of control signals between the functions of model A should be considered, rather than the transfer of information. In this regard, Yermak's approach differs from Gulenko's. It should be noted that this matter is an example of a modeling dilemma which can be solved only by making an assumption, formulating predictions based on that assumption and checking how well these predictions correspond to reality. This approach was adopted, but there is still no consensus on the issue.

An attentive reader will have noticed that the names of the functional blocks – ego, superego, id and superid – are borrowed from Freud, but this connection should be treated rather as a naming convention than a statement of full equivalence. Another issue arises due to the abstract definitions of information elements. In fact, a deep understanding of their meaning is challenging for those who are not experts in the field. Over time, the abstract definitions proposed by Augustinavičiūtė were reformulated, by herself and other thinkers. For thorough and user-friendly explanations of information elements, the reader may consult Eglit's book (Eglit, 2010). To some extent, the aspects of information represented by these elements are closely related to the areas of operation of corresponding Jungian functions, but on the other hand they have distinctive properties of their own, like statics and dynamics.

Jung was careful with his conclusions about the structure of the psyche. He emphasized the great complexity of the mechanism and the immense diversity shown in different people. In his view, traits like extraversion/introversion or conscious/unconscious should be treated as scales, in which moderate strengths of these qualities are possible, and not as simple binary oppositions. However, even if binary values are assigned to specific functions within Model A, the integrated effect of the operation of the whole system is indeed not binary, but diverse. Moreover, some initial simplifications have to be made in order to propose a fixed framework. The overall value of the model is judged based on its predictive quality. As shown in the cited works (Eglit, 2013; Horwood & Maw, 2012; Reinin, 2005), the results of its application are positive and therefore it should warrant greater interest.

The aim of socionics is not only to propose a simplified structural model of the psyche that submits to logical analysis, but also to explain misunderstandings in communication between people in terms of their typological differences and information metabolism.

These misunderstandings find logical explanations within a socionics framework. The theory of intertype relationships is broad, so it will not be presented in this paper.

6. Summary

This paper was written primarily in order to provide information on the subject of socionics to English-speaking readers. It presents the origins of the socionic model of information metabolism in the human psyche, which is largely based on the original writings of Aušra Augustinavičiūtė. Before an explanation of the rationale behind the model, the prerequisite works of Jung and Kepliński are described in brief, with a focus on the key concepts used in socionics. The implications of the structure of the psyche and information exchange as hypothesized by Augustinavičiūtė are outlined. The place of socionics in the field of personality psychology and cognitive sciences is briefly discussed. As presented in the paper, the socionic model is a convenient, systematic way to describe the dynamics of the mind. It also has predictive potential in the field of relationships between individuals with different personality types and mental diagnostics, yet a more elaborate discussion on this topic is left for another occasion.

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